

India develops cooling technology for EVs and AI systems

Developed at the Indian Institute of Technology Gandhinagar, the research team has developed a friction stir channeling (FSC) manufacturing process that simplifies liquid cold



IIT Gandhinagar has developed country's first own cooling tech for EVs, AI infra (Credit: <https://indianexpress.com>)

plate production for EVs, AI data centres, railways, and electronics by reducing leaks and manufacturing complexity. The TRL 7 technology forms cooling channels directly within a single metal plate, eliminating multiple brazed joints. The largest prototype withstood pressures above 35 bar and passed fatigue and tensile tests. The process lowers energy use, material waste, and carbon emissions while strengthening India's domestic thermal management manufacturing capabilities.

Better thermal camera accuracy without cryogenic cooling

Researchers at Yale University have developed a transistor-based technique that improves thermal camera accuracy without cryogenic cooling. The method uses a two-terminal NPN transistor to create a carrier feedback loop that increases the temperature coefficient of resistance (TCR) of conventional thermal sensing materials from about 10% to 150% per kelvin. This allows thermal sensors to detect smaller temperature changes while using existing materials instead of new detector technologies. Compatible with standard silicon manufacturing, the approach could lower the cost and complexity of thermal cameras for autonomous vehicles, drones, robotics, night vision, remote temperature monitoring, firefighting, and search-and-rescue applications.

Tiny antenna could make medical implants better

A team at the University of Glasgow has developed an ultra-small wireless antenna that could significantly reduce the size of implantable medical devices while enabling

much higher data transmission through body tissue. The sub-millimetre μ Bots use magnetoelectric antennas instead of conventional RF designs, lowering heat generation and power consumption while achieving bandwidths of up to 22.6GHz. Developed in Glasgow, the technology successfully transmitted real-time video and audio through biological tissue and maintained reliable performance in long-term simulation tests. The breakthrough could enable advanced implants for continuous health monitoring, targeted drug delivery, brain stimulation, and early diagnosis of neurological disorders such as Parkinson's disease and epilepsy.

3D-printed zinc battery holds much more charge

Researchers at UCLA have developed a zinc-ion hybrid energy storage device with a 3D-printed porous electrode that stores more than seven times the charge of comparable systems, offering a potential boost for grid-scale



A UCLA-led research team developed a 3D-printed electrode with a hollow structure that expanded the capacity of hybrid zinc-ion energy storage devices (Credit: <https://newsroom.ucla.edu>)

energy storage. Developed at the University of California, Los Angeles, the device combines battery-like energy storage with supercapacitor-style fast charging using a honeycomb carbon

electrode coated with vanadium oxide. It retained 82% capacity after 1500 cycles. The team also created a sealed 3D-printed test cell that improved measurement consistency and significantly extended electrode testing performance compared with conventional open-cell methods.

New material replaces platinum for hydrogen fuel production

Engineered by a team at the Indian Institute of Technology Guwahati, a multifunctional MXene-based material efficiently produces hydrogen through water electrolysis while enabling solar-powered seawater desalination. The ultrathin ribbon-like catalyst uses ruthenium atoms to enhance catalytic activity, achieving a hydrogen evolution overpotential of just 12mV—better than commercial platinum-carbon electrodes. Integrated into a floating Janus evaporator, it delivers a solar evaporation rate of about 3.2kg/m²/hr without salt accumulation. The technology could support sustainable hydrogen production and affordable clean water generation for energy and water-scarce regions.